Functions in C++

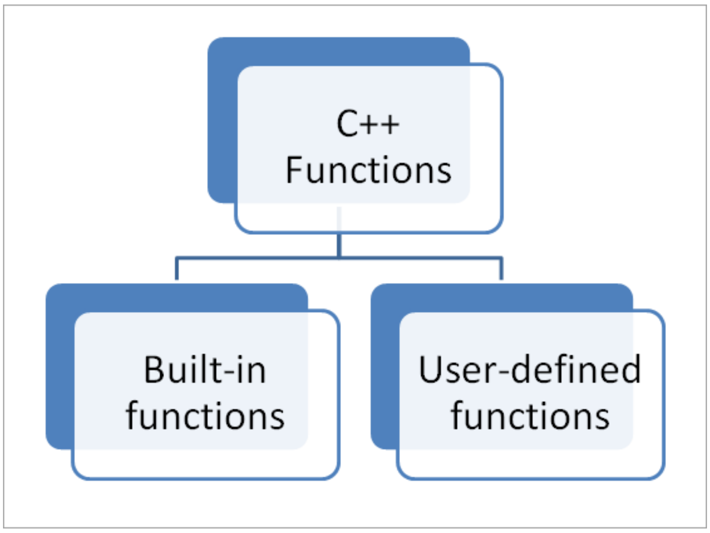
- A function is a group of statements that together perform a task. Every program in C++ has main() function by default. C++ standard library provides lot of builtin functions that you can call or use in your program.

OR

- A function is a block of code that performs a specific task and can be reuse anywhere in the program.

There are two types of function:

1. Standard Library Functions / Predefined Function / Built in Functions
2. User-defined Function / Custom Function



1. Predefine Function

- Predefine functions are the functions that are already written and stored in c++ standard libraries. Programmers can directly use them in their program without writing their definition.
- You only need to include the correct header file to use them.

OR

- A predefined function in C++ is a function that is already define in the C++ standard library and can be used directly by including the appropriate. header file.

2. User define Function

- A User defined function in C++ is a function that is define by the programmer to perform a specific task and can be called whenever needed in the program.

❖ Header Files

- Contains a collection of function prototypes, constant and preprocessor definitions
- Named with extension .h

A header file may be included in one of the two ways.

include <iostream.h> (use angle bracket <>)or
include "iostream.h". (use double quotes " ")

✓ The header file in angle brackets means that file reside in standard include directory (predefined files).

e.g.- #include <iostream>
#include <cmath>
#include <cstring>,…………

✓ The header file in double quotes means that file resides in current directory(User defined files).

e.g. - #include "hira.h"
#include "Bihar.h",…………

❖ Preprocessor

- Preprocessor commands begin with a # and do process of instruction available in #.

e.g. #include – includes a named file
#define – defines a (text replacement) macro

#ifdef/#else/#endif – conditional compilation

❖ **# define =** Often used to define constants.

e.g. #include <iostream>

#define PI 3.14 //define constant & PI (identifier) is a macro

const float PI = 3.14; // define constant

#define vs constants

#define	Constants
It has no data type No memory is allocated It is replaced by the preprocessor before compilation. Debugging is difficult.	It has a proper data type (float) Memory is allocated It is handled by the compiler. It is safer and more modern.

❖ List of Predefine Function header files in c++ and its use/work

1. <iostram>

Used for input and output operations in C++.

Name	Function Use
cout	Display output on screen {cout→it is standard output stream i.e Object, not function}
cin	Takes input form keyboard
endl	Moves cursor to next line OR New line
getline ()	Reads full line input
put ()	Writes a character
get()	Reads a character
write()	Writes block of data
read()	Reads block of data

Q. Program in C++ use all function in iostream

```
#include<iostream>
#include<conio.h>
#include<limits> // use buffer clear
using namespace std;
```

```
int main()
{
char c;
char n[20];
string m;
cout << "Enter you name = ";
cin >> c;
cout << "Your name is = " << c << endl;    // read one character
cin.ignore(numeric_limits<streamsize>::max(), '\n'); // buffer clear
include #limits
cout << "====================================" <<endl;
cout << "Enter your name again = ";
cin >> n;
cout << "Your name is = " << n << endl;
cin.ignore(numeric_limits<streamsize>::max(), '\n'); // buffer clear
cout << "====================================" <<endl;
cout << "Enter your name again = ";
getline(cin,m);
cout << "Your name is = " << m << endl;
cout << "====================================" <<endl;
char ch;
cout << "Ek character likhiye = ";
ch = cin.get();
cout << " aapne likha : ";
cout.put(ch);
   getch();
   return 0;
}
```

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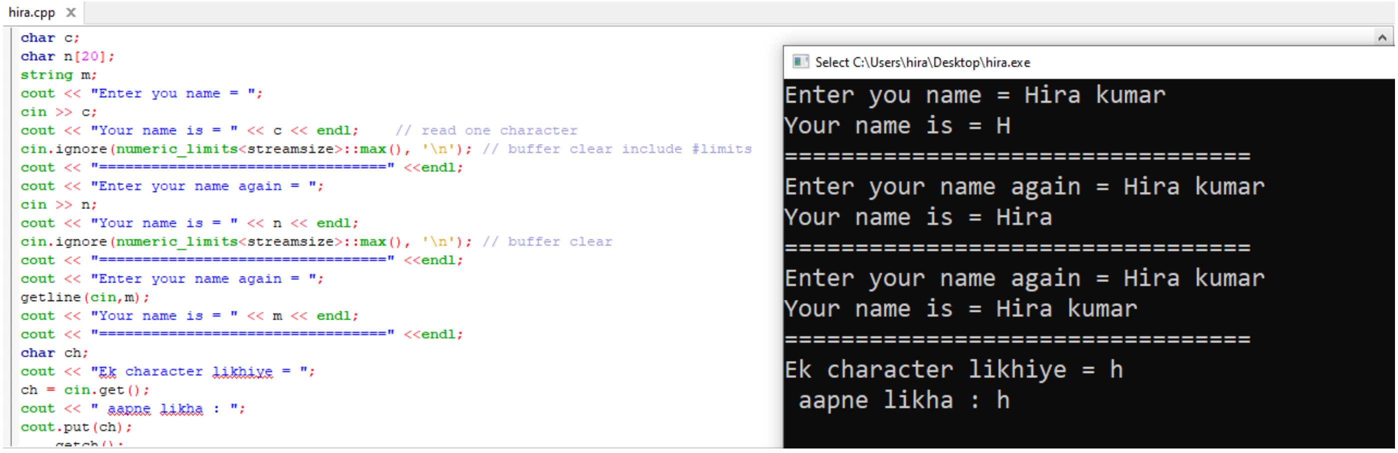


hira.cpp

```
char c;
char n[20];
string m;
cout << "Enter you name = ";
cin >> c;
cout << "Your name is = " << c << endl;     // read one character
cin.ignore(numeric_limits<streamsize>::max(), '\n'); // buffer clear include #limits
cout << "===================================" <<endl;
cout << "Enter your name again = ";
cin >> n;
cout << "Your name is = " << n << endl;
cin.ignore(numeric_limits<streamsize>::max(), '\n'); // buffer clear
cout << "===================================" <<endl;
cout << "Enter your name again = ";
getline(cin,m);
cout << "Your name is = " << m << endl;
cout << "===================================" <<endl;
char ch;
cout << "Ek character likhiye = ";
ch = cin.get();
cout << " aapne likha : ";
cout.put(ch);
```

Select C:\Users\hira\Desktop\hira.exe

```
Enter you name = Hira kumar
Your name is = H
===================================
Enter your name again = Hira kumar
Your name is = Hira
===================================
Enter your name again = Hira kumar
Your name is = Hira kumar
===================================
Ek character likhiye = h
 aapne likha : h
```

2. <cmath> or <math.h>

Used for mathematical calculations.

name	Function use
sqrt()	Square root
pow(x,y)	Power
abs(x)	Absolute value (convert -ve to +ve)
ceil(x)	Round up
floor(x)	Round down
sin(x)	Sine (x will be insert in radian value not degree) Radian = degree $\times \frac{\pi}{180}$ ($\prod$ =3.141)
cos(x)	Cosine
tan(x)	Tangent
log(x)	Natural logarithm
log10(x)	Log base 10
exp(x)	Exponential
fmod(x,y)	Remainder
hypot(x,y)	hypotenuse

Q. Write a program in c++ use all function use in cmath header file.

```
#include<iostream>
#include<conio.h>
#include<cmath>
using namespace std;
int main()
{
int x=9,a,y=2;
```

```
float aa;
a = sqrt (x);
cout << " Square root of " << x << " is " << a << endl;
cout << "==========================================" << endl;
a=pow(x,y);
cout << " square/power(2) of " << x << " is " << a << endl;
cout << "==========================================" << endl;
int z= -10;
a=abs(z);
cout << " absolute value of " << z << " is " << a << endl;
cout << "==========================================" << endl;
float c= 10.252;
a=ceil(c);
cout << " Ceil(round up) " << c << " is " << a << endl;
cout << "==========================================" << endl;
a=floor(c);
cout << " floor(round down) " << c << " is " << a << endl;
cout << "==========================================" << endl;
int d=30; // 30 in degree
float e=d*3.141/180;
aa=sin(e);
cout << " sin " << d << " is " << aa << endl;
cout << "==========================================" << endl;

   getch();
   return 0;
}
```

C:\Users\hira\Desktop\hira.exe

```
 Square root of 9 is 3
=======================================
 square/power(2) of 9 is 81
=======================================
 absolute value of -10 is 10
=======================================
 Ceil(round up) 10.252 is 11
=======================================
 floor(round down) 10.252 is 10
=======================================
 sin 30 is 0.499914
=======================================
```

3. <cstring> (character string) OR string.h

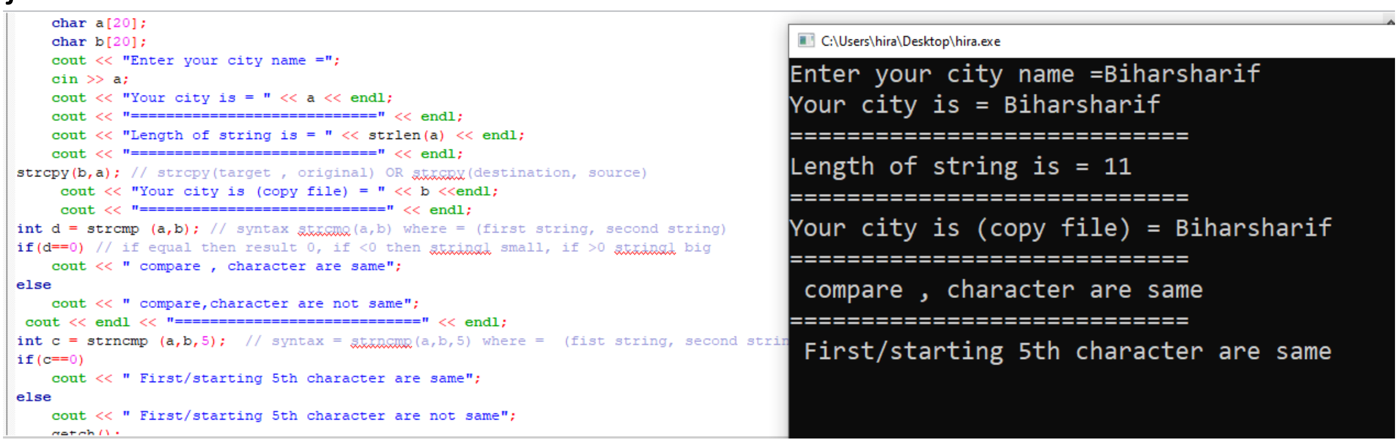
Used to wrok with text data (sequence of characters). It's much easier than C-style character arrays.

Name	Function use	syntax
Strlen()	Length of string	
Strcpy()	Copy string	
Strncpy()	Copy limited characters	
Strcmp()	Compare limited characters	
Strcat()	Concatenate or Add two two strings	
Strcat()	Concatenate limited characters	
Strchr()	Find character	
Strstr()	Find substring	
Memcpy()	Copy memory	
Memset()	Set memory	

Q. Write a program use all function available in cstring.

```
#include<iostream>
#include<conio.h>
#include<cstring>
using namespace std;
int main()
{
```

```
    char a[20];
    char b[20];
    cout << "Enter your city name =";
    cin >> a;
    cout << "Your city is = " << a << endl;
    cout << "=============================" << endl;
    cout << "Length of string is = " << strlen(a) << endl;
    cout << "=============================" << endl;
strcpy(b,a); // strcpy(target , original) OR strcpy(destination, source)
     cout << "Your city is (copy file) = " << b <<endl;
     cout << "=============================" << endl;
int d = strcmp (a,b); // syntax strcmo(a,b) where = (first string, second string)
if(d==0) // if equal then result 0, if <0 then string1 small, if >0 string1 big
    cout << " compare , character are same";
else
    cout << " compare,character are not same";
 cout << endl << "=============================" << endl;
int c = strncmp (a,b,5);  // syntax = strncmp(a,b,5) where =  (fist string, second string, compare till how many character)
if(c==0)
    cout << " First/starting 5th character are same";
else
    cout << " First/starting 5th character are not same";
    getch();
    return 0;
}
```



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Q. Write a program use all function available in cstring.

```
#include<iostream>
#include<conio.h>
#include<cstring>
#include<cstdlib> // clear screen
using namespace std;
int main()
{
   char a[20] = "Hello ";
   char b[20]= "Good";
   char c[20]= "Morning";
   strcat(a,b);
   cout << " Add = " << a << endl;
   cout << "=============================" << endl;
   strncat(b,c,5); // kitna character add krna hai
   cout << "Add = " << a <<endl;
   cout << "=============================" << endl;
   char *z =strchr(a,'o'); // pointer ka use krke search hoga
   if (*z=='o')
      cout << " character found";
   else
      cout << "Character not found";


   getch();
   return 0;
}
```

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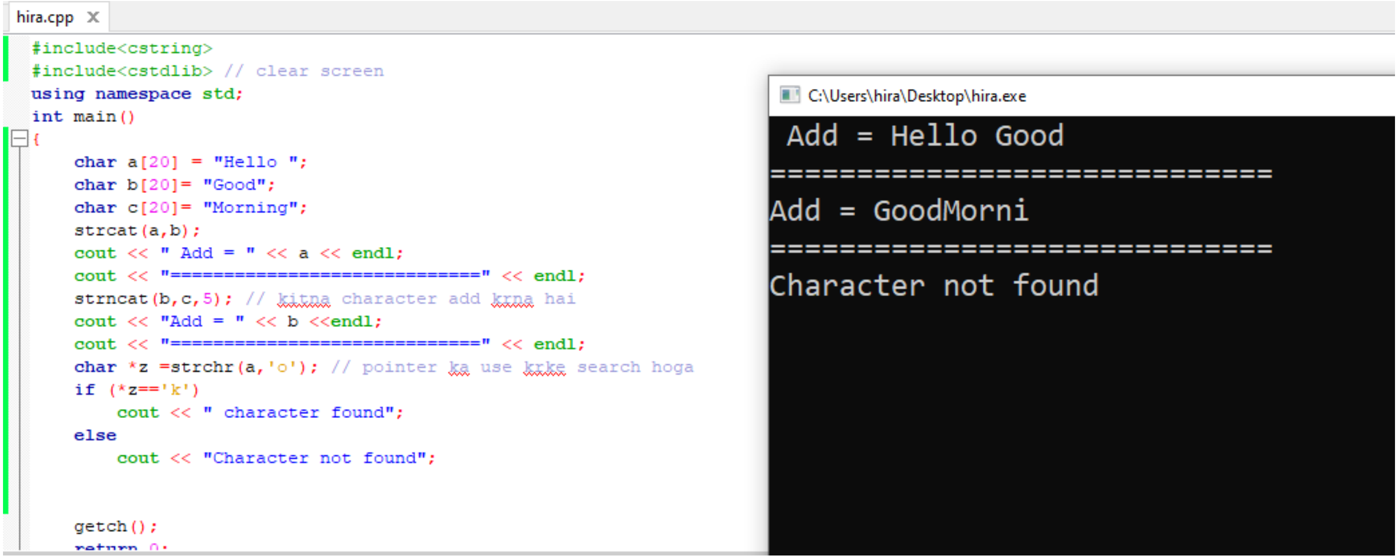
Run again



4. <cctype>

Used for character testing and conversion.

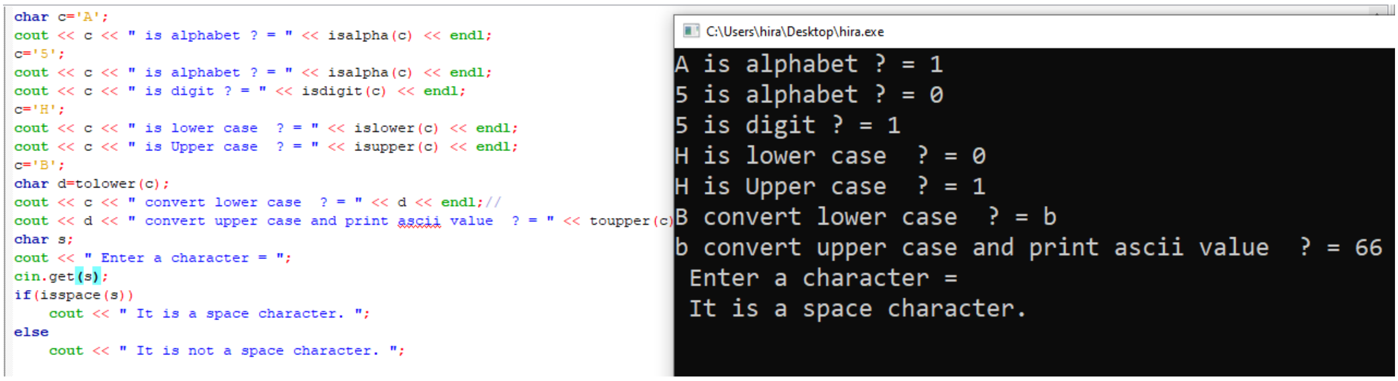
name	Function use	syntax
isalpha()	Albhabet check	
isdigit()	Digit check	
isalnum()	Alphanumeric check	
islower()	Lower case check	
isupper()	Upper case check	
isspace()	Space check	
toupper()	Convert to uppercase	
tolower()	Convert to lowercase	

Q. Write a program to print all function available in cctype header file.

```
#include<iostream>
#include<conio.h>
#include<cctype>
//#include<cstdlib> clear screen
using namespace std;
int main()
{
char c='A';
cout << c << " is alphabet ? = " << isalpha(c) << endl;
```

```
c='5';
cout << c << " is alphabet ? = " << isalpha(c) << endl;
cout << c << " is digit ? = " << isdigit(c) << endl;
c='H';
cout << c << " is lower case  ? = " << islower(c) << endl;
cout << c << " is Upper case  ? = " << isupper(c) << endl;
c='B';
char d=tolower(c);
cout << c << " convert lower case  ? = " << d << endl;//
cout << d << " convert upper case and print ascii value  ? = " <<
toupper(c) << endl;
char s;
cout << " Enter a character = ";
cin.get(s);
if(isspace(s))
   cout << " It is a space character. ";
else
   cout << " It is not a space character. ";

   getch();
   return 0;
}
```



5. <fstream>

Used for file handling.

Class	Use
ofstream	In file write (data insert)
ifstream	File read (get/collect data)
fstream	Read & Write both

name	Function use	syntax
open()	Open file	
close()	Close file	
read()	Read from file	
write()	Write to file	
getline()	Read full line	
eof()	End of file check	
seekg()	Move get pointer	
seekp()	Move put pointer	
tellg()	Get position	
tellp()	Get position	

Q. Write a program in c++ create a file or open a file and insert data.

*hira.cpp

```
#include<iostream>
#include<conio.h>
#include<fstream>
//#include<cstdlib> clear screen
using namespace std;
int main()
{
    ofstream hira; // ofstream is class & hira is a object
    hira.open("y.txt"); // file open OR file create
    hira<<"hello z"; // data insert in file
    hira.close();

    getch();
    return 0;
}
```

Q. Write a program in c++ open a file and read data.

hira.cpp

```
#include<iostream>
#include<conio.h>
#include<fstream>
//#include<cstdlib> clear screen
using namespace std;
int main()
{
  /* string a;
  cout << "Enter your name =";
  getline (cin,a);
  cout << "Your name is = " << a; */
    ifstream hira;
    hira.open("y.txt");
    string s; // y.txt file data store/copy in datatype s variable
    hira>> s;
   cout <<s << endl;

    hira.close();

    getch();
    return 0;
}
```

J:\.apk\hira.exe

```
hello_Good_Morning_India
```

y - Notepad

File Edit Format View Help

```
hello_Good_Morning_India
```

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6. <algorithm>

Used for sorting and searching algorithms.

7. <string>

Used to work with string class.

8. <vector>

Used for dynamic arrys.

9. <iomanip>

Used for formatted output. E.g. – alignment,fixed decimal,….

10. <cstdlib>

Used for utility functions. E.g – exit(), free(), rand(),malloc(),…

And so many,………………………